Elizabeth H. Childs

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Mechanical engineer and HCI researcher who takes tangible, haptic, and AR interfaces from concept through hardware/software prototype to evidence-based user evaluation.

Education

Stanford University — *Stanford, CA*
PhD, Mechanical Engineering (Human-Computer Interaction Research), GPA 3.9/4.0 — *Expected Dec 2027*
MS, Mechanical Engineering (Controls); MA, Education — *May 2026*
Knight-Hennessy Scholar: Stanford's flagship graduate leadership fellowship (<1% acceptance rate)
Impact Labs PhD Fellow: competitive fellowship for solutions-oriented research with practitioners

University of Maryland, College Park — *College Park, MD*
B.S., Mechanical Engineering, GPA 4.0/4.0 — *Dec 2020*
Banneker/Key Scholar: UMD's most selective merit scholarship (full tuition, room, and board)

Research & Industry Experience

Stanford University, Stanford HCI Group & CHARM Lab — *Stanford, CA*
PhD Researcher (Advisors: James Landay, Allison Okamura) — *June 2021 – Present*
- Research tangible and AR interfaces for hands-on STEM learning, owning projects from needfinding and concept through prototyping, user studies, and publication.
- Defined the research direction by surveying 58 teachers, interviewing 18, and observing classrooms across 7 schools; findings showed AR should augment physical labs rather than replace them.
- Designed and evaluated an AR physics lab overlaid on a real pendulum; in a 52-student between-subjects study it improved scientific reasoning vs. a traditional lab ($d = 0.63$).
- Deployed an AR biology app (Unity, OpenCV) at 5 institutions with 100+ students; it tracks students' real balloons with computer vision and animates the enzyme activity behind them.
- Implemented the AR features for a 10-person team's narrative outdoor-learning app, evaluated in a field study with 44 children (ACM CHI 2023).

Microsoft Research, Future Experiences Group — *Redmond, WA*
Research Intern (Mentors: Sumit Basu, Scott Saponas) — *June 2026 – Sep 2026*
- Conceived, designed, and built a free-moving haptic robot that provides multidirectional forces and torques so learners can feel physics simulations, from airfoil lift to molecular bonds.
- Engineered a gearless omni-wheel drivetrain with custom motor control, packaging a Raspberry Pi, onboard camera, battery, and buck/boost power electronics into one compact 3D-printed body.
- Fused overhead computer-vision tracking with motor feedback for hybrid localization, keeping force rendering precise between camera frames; connected the robot to projected-AR physics simulations.

Dolby Laboratories, Advanced Technology Group — *Sunnyvale, CA*
Research Intern — *May 2024 – Aug 2024*
- Conceived and prototyped a hybrid microscope whose physical focus, magnification, and stage controls drive high-resolution imagery on a connected smartphone; presented the concept to ATG leadership.

Lam Research, Global Products Engineering — *Fremont, CA*
Mechanical Engineering Intern — *May 2021 – Aug 2021*
- Programmed HoloLens AR applications (Unity, C#) that render a live hologram of industrial robots running inside opaque enclosures, turning telemetry into spatial views for engineers.
- Pitched and delivered a tool that replays robot motion leading up to a fault, helping technicians better diagnose errors and demonstrating how AR can improve robot servicing.

University of Maryland, GAMMA Lab — *College Park, MD*
Research Assistant (Advisor: Dinesh Manocha) — *Aug 2020 – Aug 2021*
- Co-led (co-first author) an IEEE TVCG literature survey of how AR/VR can support distance learning.

Mechanical Engineering & Robotics Experience

University of Maryland, Bioinspired Advanced Manufacturing Lab *College Park, MD*
Undergraduate Researcher (Advisor: Ryan Sochol) *Sept 2018 – July 2020*

- Invented an additive assembly method for multi-material 3D-printed microfluidics (open halves with printed fasteners and seals), cutting post-processing time ~98% at equal or higher burst pressure.
- Led a 3-person team (1 undergraduate, 1 master's student) through first-author publication.

University of Maryland, Bioinspired Robotics: Flic-Flac Spider Robot *Feb 2019 – May 2019*

- Integrated 4 coordinated servos, a Teensy, and IMU orientation sensing into a somersaulting robot inspired by the flic-flac spider; it covered the course 20× faster than required, fastest of 13 teams.

Key Technologies, Inc. *Baltimore, MD*
Mechanical Engineering Intern, Medical Device Consulting *Sept 2019 – Dec 2019*

- Fabricated medical device prototypes, and designed and built test rigs for microfluidic devices.

Institute of Technology of Cambodia *Phnom Penh, Cambodia*
NSF International Research Experience for Students (Advisor: Garrett Clayton) *June 2019 – Aug 2019*

- Developed a compact modular robot to reach unexploded ordnance in dense forest.

Oregon State University, Dynamic Robotics Lab *Corvallis, OR*
NSF REU Researcher (Advisor: Jonathan Hurst) *June 2018 – Aug 2018*

- Cut peak force 47% and step impact 90% on the Cassie bipedal robot, with no controller changes, by designing impact-absorbing feet and force-plate testing 3 geometries in silicones, foams, and gels.

Selected Publications

1. **E. Childs***, F. Mohammad*, L. Stevens* et al., "An Overview of Enhancing Distance Learning Through Emerging Augmented and Virtual Reality Technologies," *IEEE TVCG*, 2024. doi:10.1109/TVCG.2023.3264577
2. A. Cheng, . . . , **E. Childs**, et al., "Designing Immersive, Narrative-Based Interfaces to Guide Outdoor Learning," *ACM CHI*, 2023. doi:10.1145/3544548.3581365
3. U. Bhattacharya, **E. Childs**, et al., "Speech2AffectiveGestures: Synthesizing Co-Speech Gestures with Generative Adversarial Affective Expression Learning," *ACM Multimedia*, 2021. doi:10.1145/3474085.3475223
4. **E. Childs** et al., "Additive Assembly for PolyJet-Based Multi-Material 3D Printed Microfluidics," *J. MEMS*, 2020. doi:10.1109/JMEMS.2020.3003858

Talks, Panels & Outreach

Poster Presentation, "ExpertLens: Bridging Physical Observations With Scientific Models through Computer Vision and AR," ACM UIST 2026
Outreach Tabling, Society of Women Engineers Conference, encouraging women toward grad school 2026
Top Poster & Talk, poster selected for a presentation, Augmented World Expo, Long Beach 2026
Panelist, Q&A for prospective engineering PhD students 2026
Workshop Co-Organizer, Purposeful XR: Ethical Futures, ACM CHI, Yokohama 2025
Panelist, The Educational Rift in Spatial Reasoning, Augmented World Expo, Long Beach 2024
Panel Moderator & Demo Presenter, XR in Education, Stanford XR Conference 2022–23

Honors & Awards

Learning Design Challenge Winner, $10.5K in total prize funding 2026
1st Place, Best Use of Looking Glass, MIT Reality Hack (90+ teams) 2024
Academic Achievement Award, highest GPA of ~300 UMD mechanical engineers 2019
NSF Bridge Scholar, selective NSF-funded program for incoming UMD students 2016

Skills

Research Methods: needfinding, interviews, surveys, qualitative coding, experiments, Bayesian analysis
Hardware: SolidWorks, 3D printing (FDM, PolyJet), CNC machining, laser cutting, PDMS (silicone) manufacturing, mechatronics, motor control, power electronics, sensor fusion, embedded systems
Software & XR: Unity, C#, C++, computer vision (OpenCV), ML, HoloLens, AR/VR, Maya